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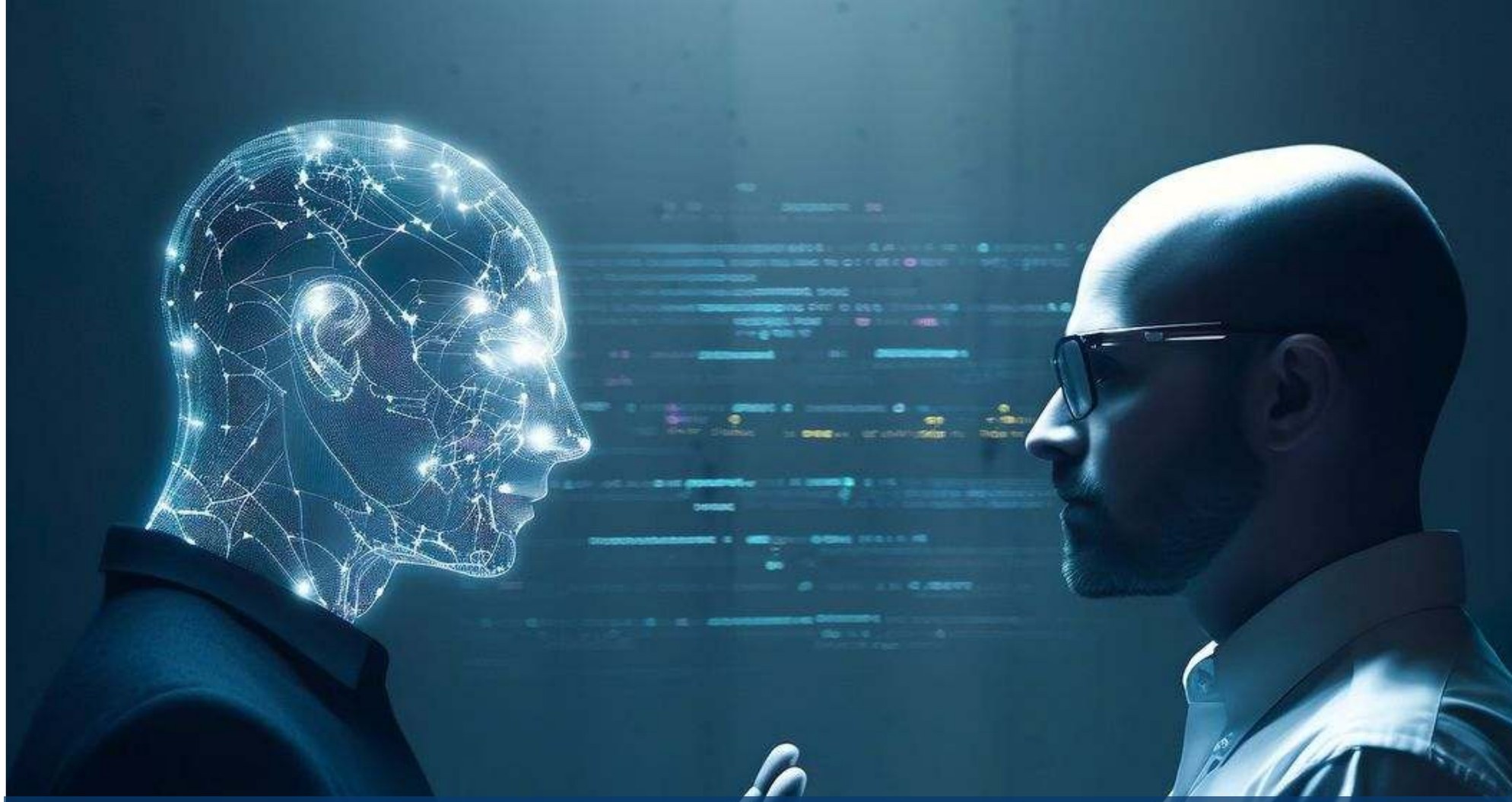

Deep Neural Networks Are More Accurate Than Humans at Detecting Sexual Orientation From Facial Images

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Research Objectives

Deep neural networks are more accurate than humans at detecting sexual orientation from facial images

- Examine a DNN-based model
- Compare the accuracy of the algorithm with human judges
- Identify which facial features contribute to prediction accuracy
- Examine whether the facial features of gay individuals show gender-atypical patterns
- Evaluate the privacy and ethical implications

Theoretical basis & Software

Deep Neural Networks (DNN)

- consist of numerous hidden layers that mimic the neocortex
- enabling them to recognize patterns in large, unstructured data
- uncover connections between characteristics and facial features overlooked by the human brain

Prenatal Hormone Theory (PHT)

- same-gender sexual orientation stems from the underexposure of male fetuses or overexposure of female fetuses to androgens
- gay men have more feminine facial features than heterosexual men, and the opposite is true for lesbians

Face-detection software

Face++

VGG-face

Study 1a:
A DNN can be used to identify sexual orientation from facial images

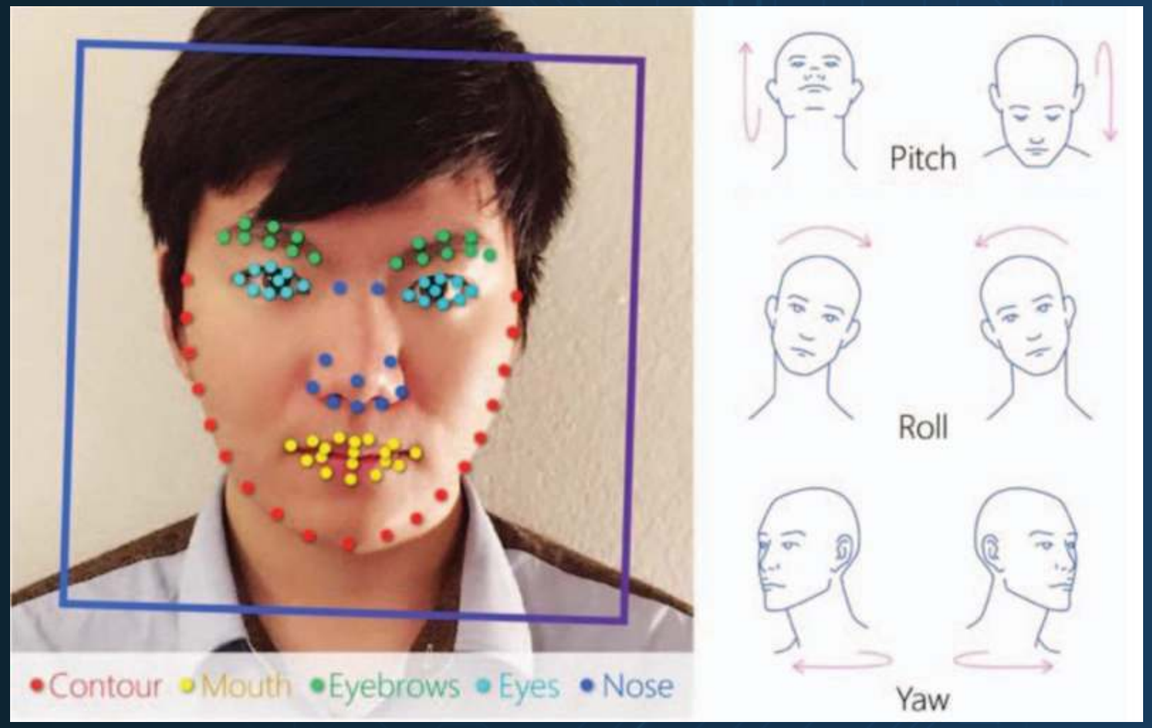


Figure 1. Graphical illustration of the outcome produced by Face++

Data Collection

- 130,741 images of 36,630 men
- 170,360 images of 38,593 women

Filtration Process

- Face++: analyze the location of the face in the image and estimate the head's pitch, roll, and yaw to remove invalid images
- Amazon Mechanical Turk (AMT) workers: verify that the faces were adult, Caucasian, fully visible, and matched with the user's profile.

Table 1. The final sample contained 35,326 facial images

	Men		Women	
Variable	Gay	Heterosexual	Lesbian	Heterosexual
Unique users	3,947	3,947	3,441	3,441
Median age (IQR)	33 (30–36)	33 (30–36)	29 (25–34)	29 (25–34)
Total images	8,996	8,645	7,457	10,228
Users with at least:				
1 image	3,947	3,947	3,441	3,441
2 images	2,438	2,439	2,037	2,878
3 images	1,363	1,367	1,058	1,951
4 images	562	731	494	1,114
5 images	219	327	223	491

Research Methodology & Results

Analysis Process:

- VGG-Face was utilized to extract 4,096 features for a facial image
- Simple prediction model, trained logistic regression, combined with singular value decomposition to train the classifiers

Result

- Men: 81% of cases with just 1 image → 91% for five images
- Women: 71% (one image) → 83% (five images per person)

Figure 2. The accuracy of the DNN-based sexual orientation classifier against the number of images used in the classification.

Research Methodology & Results

Study 1b: Elements of the Facial Image Employed by the Classifier

Data Collection

- Randomly drawn facial images of 100 males and 100 females from study 1a
- Adjusted the faces in exactly the same direction for all images

Classifiers

- Used the remaining images to train separate sexual orientation classifiers for men and women.

Analysis Process

- Used the trained classifiers to estimate the probability of being gay for the faces
- Covered the top left of all 100 images and estimated the probability again
- Repeated 1,024 times and measured the changes in predictions.

Research Methodology & Results

Figure 3. Heat maps depict the extent to which masking specific areas of an image affects the overall classification result

Note: blue (no change) to red (substantial change)

Result

- For both genders, classification mainly relied on the facial area and ignored the background
- The most informative facial areas among
  + Men: nose, eyes, eyebrows, cheeks, hairline, and chin
  + Women: nose, mouth corners, hair, and neckline

VGG-Face scores extracted here focus on the facial features rather than on other parts

Study 1c

The difference between gay and heterosexual faces that enable the classification

Data Collection

- Utilized facial images and probabilities from Study 1a, focusing on faces directly facing the camera (pitch and yaw within two degrees)
- Selected subsets of images classified as most likely and least likely to be gay
- Generated average landmarks' locations using 500 images per subset
- Created composite faces using 100 images per subset

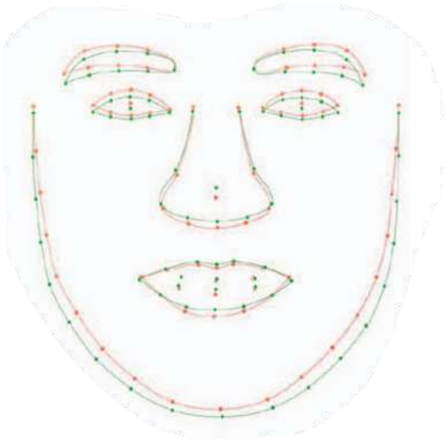
Result

The findings support the Prenatal Hormone Theory (PHT), which suggests gay faces tended to be gender atypical

- For men:

  + Average landmark locations revealed that gay men had narrower jaws, longer noses, and larger foreheads

  + Also, gay men had less facial hair and lighter skin

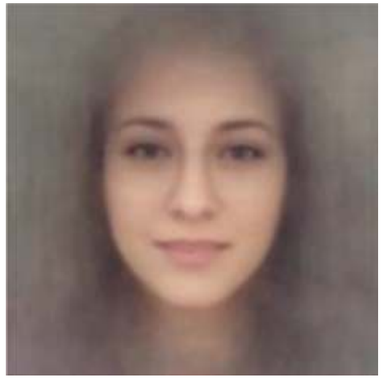


Gay

Straight

Result

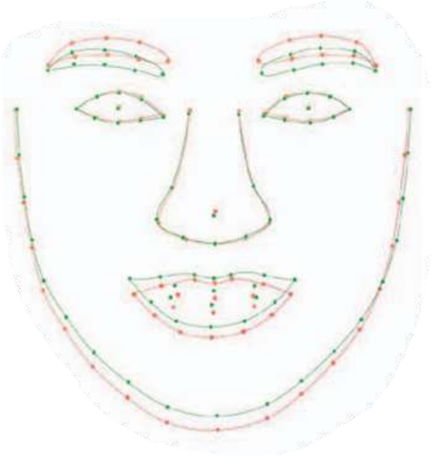
The findings support the Prenatal Hormone Theory (PHT), which suggests gay faces tended to be gender atypical



- For women:

  + Average landmark locations revealed that lesbians had larger jaws and smaller forehead

  + Lesbians tended to use less eye makeup, had darker hair, and smiled less than their heterosexual counterparts



Lesbian

Straight

Research Methodology & Results

Study 2: The faces of gay men and women are "gender-atypical"

Classifier

DNN-based gender classifier: was developed on an independent sample of 2,891,355 facial images

Analysis Process:

Used the same approach to Study 1a: facial images and trained the classifier

Result

The faces of gay men were more feminine and the faces of lesbians were more masculine

The accuracy based on one facial image: AUC .57 for men and AUC .58 for women

Study 3: The basic nontransient morphological features provide enough information to classify sexual orientation

By using the same sample in Study 1a and Face++ to extract morphological features (83 landmarks), the result showed that:

- The shape of individual facial elements enabled high classification accuracy for both genders
- Facial contour provided remarkably high accuracy: 75% for men and 63% for women.
- The outline of facial features like eyes, eyebrows, and mouth is influenced by expression, but facial contour remains relatively stable.

Study 3: The basic nontransient morphological features provide enough information to classify sexual orientation

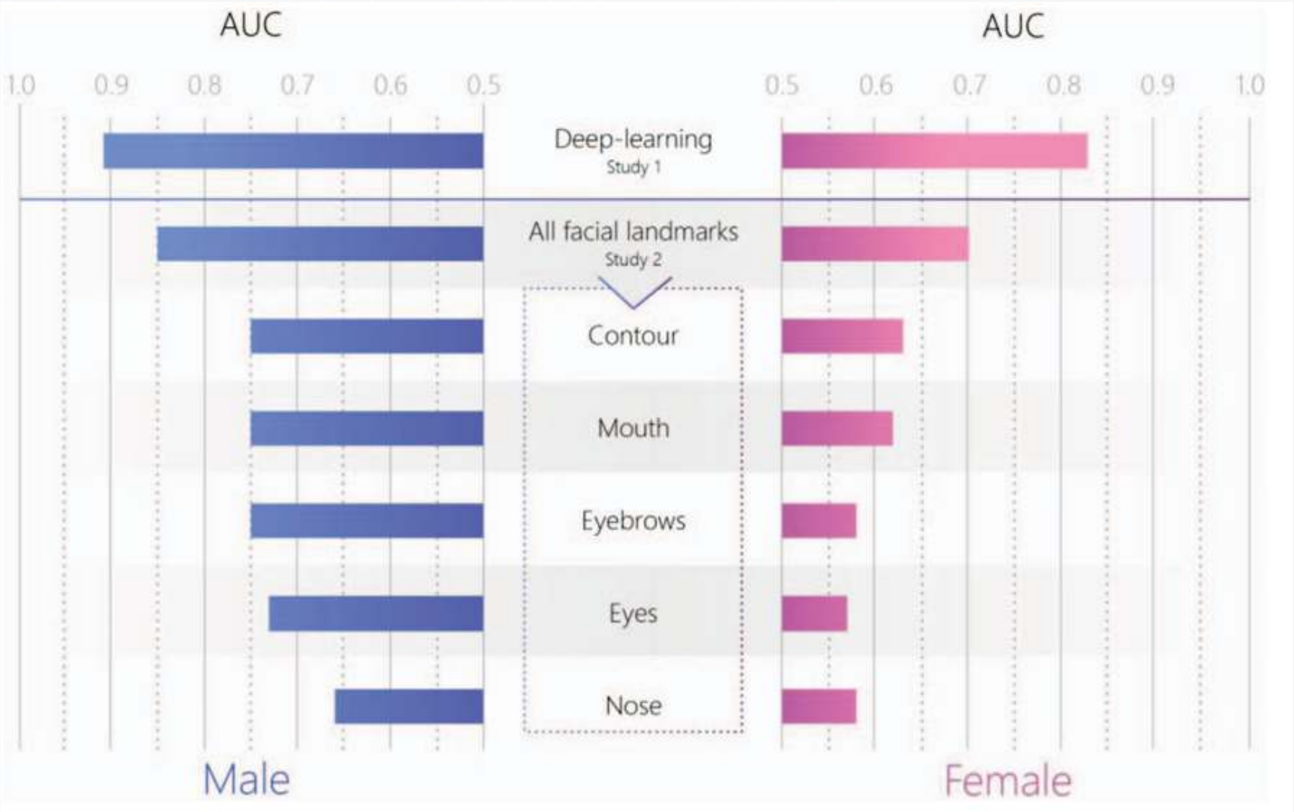


Figure 5. The accuracy of the landmark-based classifiers (5 images / person)

Research Methodology & Results

Study 4: Human Judges

Is it possible that facial images posted on a dating website are revealing of sexual orientation?

Method

AMT workers were asked to select the facial image more likely to represent a gay person from two, randomly ordered, facial images of the sample in Study 1a

Result

Human judges achieved an accuracy of AUC .61 for male images and AUC .54 for female images

Study 5: Beyond Dating Website Facial Images

Data collection

14,438 facial images of 6,075 openly gay men from the myPersonality database

Method

Face images and VGG-Face as in Study 1a

Final Sample contained 918 facial images

Result

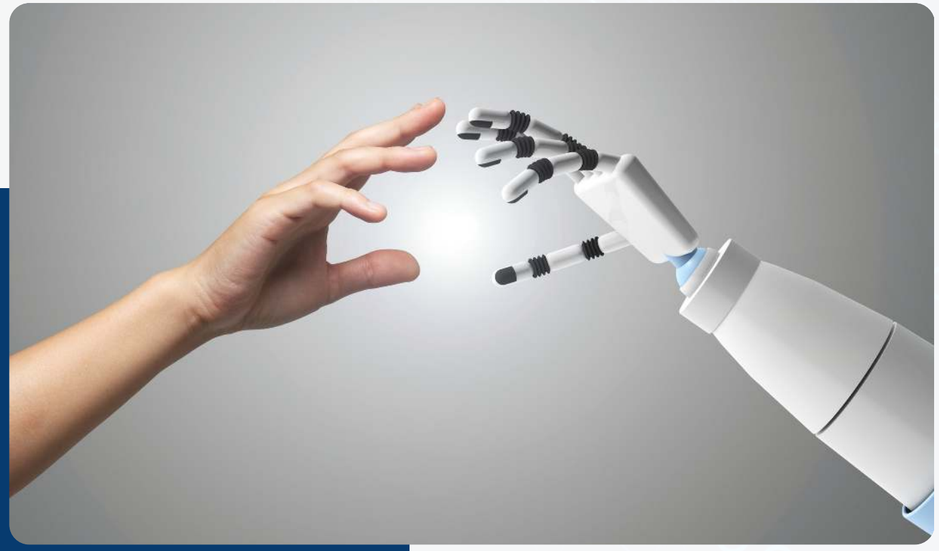
The classifier could accurately distinguish between:

+ Gay Facebook users and heterosexual dating-website users in 74% of cases
+ Gay Facebook users and gay dating-website users in 53% of cases

Main Contributions & Conclusions

- Deep Neural Networks Are More Accurate Than Humans at Detecting Sexual Orientation From Facial Images
- Predictions rely on information within the facial region, not background cues
- Gay individuals show gender-atypical facial characteristics, supporting the Prenatal Hormone Theory

Main Contributions & Conclusions



- The strong performance of classifiers based solely on fixed morphological features
- Human judges remain consistently inaccurate, even when evaluating nonstandardized images
- The classifier maintains similar accuracy when applied to images collected in different environments

Limitations

Used Non-standardized images: characterized by varying quality, head orientation, or facial expression

Pro: higher ecological validity & a more representative sample

Con: Introduces confounders

Openly gay users may differ than ones that have not “come out“

The photos acquired from a dating site potentially could be taken to be more revealing of their orientation.

Some users could have miscategorized themsevles

Only White participants from the U.S.

To address limitations

- The researchers employed a model that was specifically trained to focus on nontransient facial features.
- Focusing on faces that were not substantially more revealing of their sexual orientation.

Implications

Scientific implication

This study could serve as a foundation and reference for future research

Ethical and Social Implications

This study serves as a wake-up call, urging everyone to be more careful with their personal image on social media platforms.

Policy Implications

- Policymakers need transparent policies on image storage and processing, data security, and raising public awareness of potential risks
- Enforce the creation of legislation to help protect the general public and the homosexual communities across the world

THANKS FOR LISTENING